Robotics Setter Technician Program

Follow-up 2 | October 26, 2026

Diego Adair de León Márquez

Universidad Politécnica de Aguascalientes | Mechatronics Engineering, 9th term
Plant Tutor: Fernando Robledo
Engineering Department | August 1, 2026 to January 1, 2027

TENNECO

TENNECO CONFIDENTIAL

THE TENNECO VALUE I IDENTIFY WITH

Win

We must earn the trust of our employees and customers.

For me, winning means caring for the ergonomics of our welders while protecting product quality. As an engineer, I do it by optimizing the process with robotic systems.

Program scope and objective

Robotics Setter Technician | August 1, 2026 to January 1, 2027

OBJECTIVE

Become a specialist in robotic welding and advanced technical support.

WHY IT MATTERS

When a welding robot stops, the line stops with it. A setter who can find the root cause and adjust the cell on the spot keeps parts moving and quality under control.

FINAL PROJECT

An implemented improvement

In quality, availability or productivity.

Mine: 4M, implemented, with 47% less cycle time.

An updated training process

How the next setters learn.

Mine: study material reviewed, schedule rebuilt.

The plan: four levels, each one validated

My ILUO matrix, paraphrased | 40 hours of formal training

OBJECTIVE Become a specialist in robotic welding and advanced technical support.

LEVEL		WHAT IT COVERS	HOW IT IS VALIDATED	TIME	METHOD
I	**Understands**	Safety, GMAW equipment, metal transfer, process variables, discontinuities, joints, positions, symbols and control documents.	Theory exam, 8.0 to pass	**4 h**	Classroom session
L	**With support**	Basic robotic and dimensional adjustments, checking settings against the WPS, finding defects and changing consumables.	Validation on the floor	**8 h**	Robotic cell, on the floor (Gemba)
U	**Alone**	Creating new welding programs in SKS and setting their parameters (SKS and Miller).	Practical exam	**8 h**	Robotic cell, on the floor (Gemba)
O	**Improves and teaches**	Training level 2 and 3 staff, improvement ideas, problem solving (8D, Ishikawa, 5W+2H), GD&T, ANDON and Lean.	Implementation and training, validated	**20 h**	80-hour follow-up: TM, WAVE or PSIF

TENNECO

The plan slipped where the floor needed me

Since the last review | Commitments from September 25

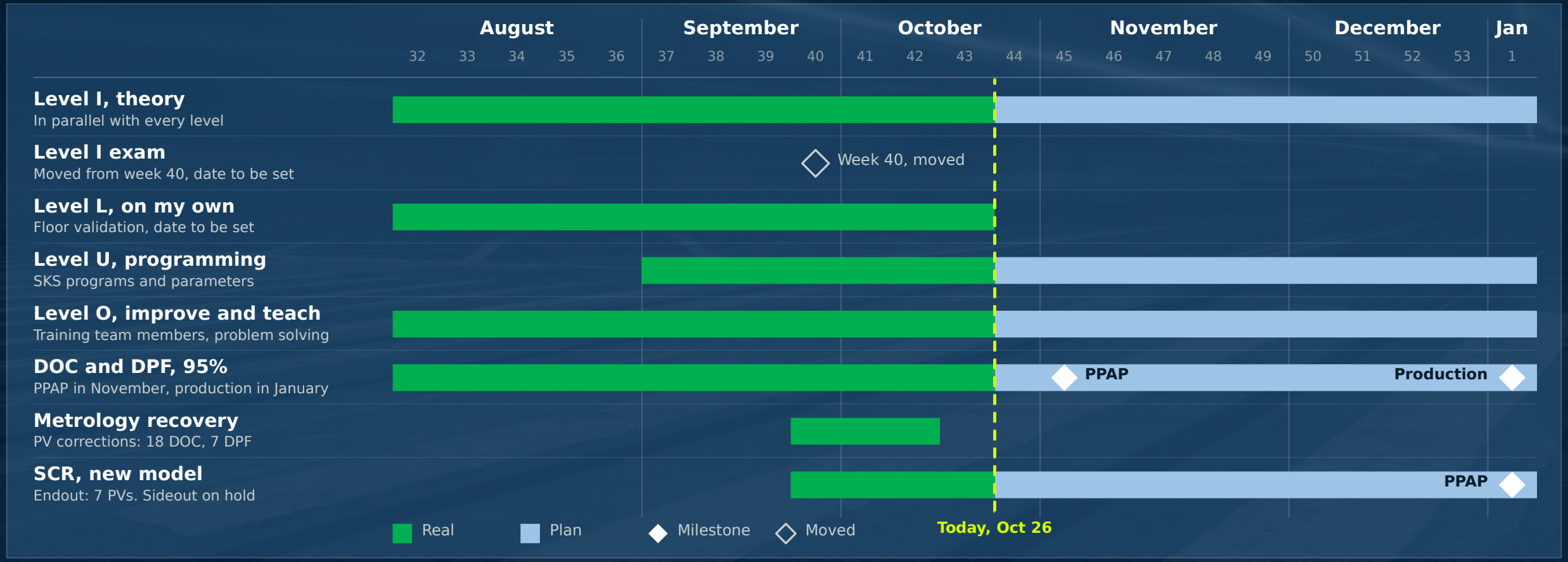
WHAT I COMMITTED TO	STATE	WHAT HAPPENED
Close the four open Level I topics	Done	Closed. Theory keeps growing in parallel with the other levels.
Pass the Level I exam in week 40	Late	Not taken yet. Date to be set.
Validate Level L on the floor in October	Late	I already do all five skills on my own. Date to be set.
Reach Levels U and O before January	On track	All three U skills and four of six O skills are moving.
Keep measuring 4M and find the next improvement	On track	4M ran the DOC and DPF PV parts. The search continues.

WHY

In October the industrialization project needed the whole team: parts already released failed metrology, and the SCR model started. The exams waited; the floor did not.

Skills kept moving while the exams waited

Gantt | Technical and practical skills, done and next



TENNECO

15 of 23 skills achieved, only 2 not started

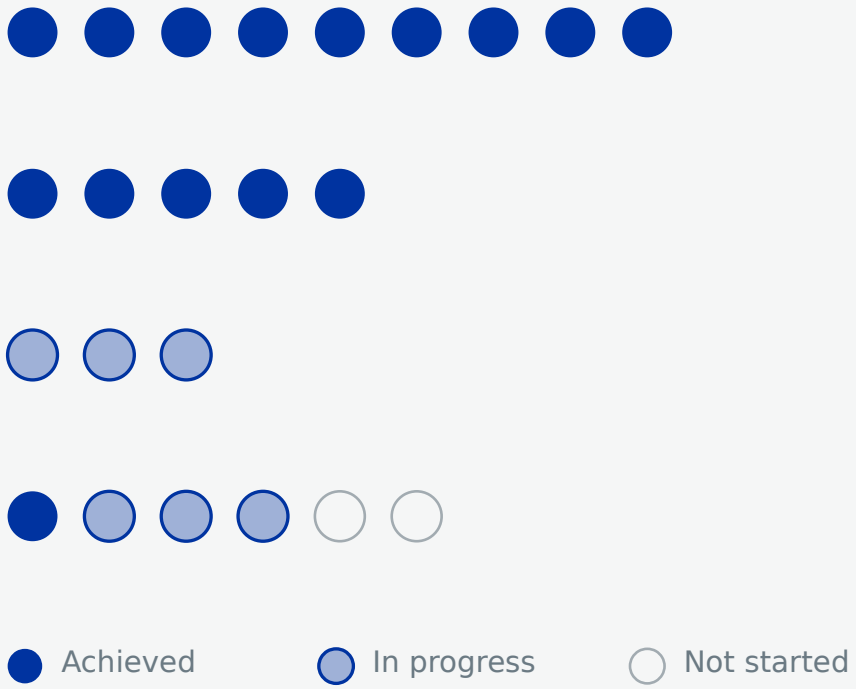
ILUO status | September 25: 11 achieved, 6 not started

I **Understands**
9 skills

L **With support**
5 skills

U **Alone**
3 skills

O **Improves**
6 skills



WHAT MOVED

I Four topics closed; theory keeps growing

L All five skills, now on my own

U SKS programs and parameters started

O Training team members, problem-solving basics

Still pending: the Level I exam and the Level L floor validation.

October’s firefight taught me more than the plan

What I learned | Weeks 40 to 42

WHAT HAPPENED

DOC and DPF parts that were already released started failing metrology. A metrology expert measured everything again: the main failure was parallelism.

At the same time the SCR model started: bigger, heavier and harder to build. One of its two models, sideout, is on hold for metrology too.

WHAT I LEARNED ON THE FLOOR

- Dimensional adjustments on welding fixtures
- Tuning welding parameters until the part is right
- Reviewing the process step by step
- Training team members on machines, adjustments, process and product
- The basics of problem-solving tools

HOW On the floor, alongside Engineering and the metrology expert, while the line kept running.

TENNECO

Industrialization: DOC and DPF go to PPAP in November

Project | MY27 Cummins program, together with Engineering

THE PROBLEM

DOC and DPF parts already released failed metrology. The main failure: parallelism.

WHAT WE DID

A metrology expert measured everything again, and we corrected 25 PV parts: 18 DOC, 3 long and 4 short canning.

WHERE IT STANDS

DOC and DPF at 95%: PPAP in November, production in January. SCR: 7 endout parts built, sideout on hold, PPAP in January.

PHOTO

DOC and DPF parts, or the metrology check

MY TIME THIS MONTH

70% **industrialization, with Engineering**

30% support at the robotic cells

DOC and DPF moved from 90% to 95%

Results | September 25 to October 26

DOC AND DPF, READY TO LAUNCH



Only documentation left. PPAP in November, production in January.

17

PV parts in 4M

9 DOC, 4 DPF short, 4 DPF long

25

PV parts corrected

18 DOC, 3 DPF long, 4 DPF short

7

SCR endout parts

sideout on hold for metrology

4M, STILL RUNNING

93 s

per part in the robotic cell

−47%

cycle time, from 174 s welding by hand

Data: engineering team, October 2026

TENNECO

Next: set the exam dates, then Level U

Next steps | Until January

WHAT COMES NEXT

Level I exam and Level L floor validation

Dates to be set with my supervisor

Level U: SKS programs and welding parameters

November to December

Level O: problem solving and the next improvement

November to December

PPAP: DOC and DPF in November, SCR in January

DOC and DPF production starts in January

WHAT I NEED

- Dates for my Level I exam and Level L validation
- Supervised time programming in SKS and Miller
- Feedback on the training proposal

Conclusions and next steps

What is in place, and what comes next

IN PLACE

15 of 23 skills achieved; Levels I and L done on my own.

DOC and DPF at 95%, with 25 PV parts corrected after metrology.

A month of real problem solving: fixtures, parameters and training.

NEXT

Set and pass the Level I exam and the Level L validation.

Reach Level U before January.

Support the DOC and DPF PPAP in November, and SCR’s in January.

The plan slipped a few weeks; my skills did not.

TENNECO



Thank you

Diego Adair de León Márquez | Engineering Department